Hafsa Kabir Ahmad, PhD

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Summary

I am an experienced university lecturer with vast experience in teaching, guiding, and supervising students at all academic levels. Over the years, I've effectively coordinated academic programs and offered specialized advice to help students reach their full academic potential. Furthermore, I am a seasoned data scientist who specializes in Recommender Systems in Online Education and Graph Representation Learning. My PhD in Educational Computing focused on applying Data Science to enhance and improve learners' learning experiences. My study focused on using Artificial Intelligence (AI) to change the way problems are tackled, particularly in the context of education and individualized learning. This is especially crucial in Africa, where there is a big educational issue, as the continent has a large number of out-of-school children in the world.

Education

Shaanxi Normal University *China*

PhD Computer Software and Theory *2019 — 2023*

Thesis: Recommendation Methods in MOOCs using Graph Representation Learning.

Bayero University Kano *Nigeria*

MSc Computer Science *2012 — 2015*

Dissertation: Design and Implementation of Mobile Learning Application: A Case Study of Department of Computer Science, Bayero University Kano.

Bayero University Kano *Nigeria*

BSc Computer Science *2006 — 2011*

Work Experience

Bayero University Kano *Nigeria*

Latest Position: Senior Lecturer *April. 10 2012 — To Date*

Faculty of Computing, Bayero University Kano, Gwarzo Road, Kano 700241, Nigeria.

Teaching

Bayero University Kano

Advanced Computer Programming	*Postgraduate*
Advanced Topics in Computer Science	*Postgraduate*
Research Methodology	*Undergraduate*
Special Topics in COmputer Science	*Undergraduate*
Object-Oriented Programming	*Undergraduate*
Programming in C++	*Undergraduate*
Multimedia techniques	*Undergraduate*

Administrative Positions

2012 to 2018	Level Coordinator
2024	Postgraduate Coordinator
2026	Faculty Deputy Dean

Research Interests

- Artificial Intelligence, Educational Data Mining,GRL, ,Personalised Learning System

Skills

Technical Proficiencies:

- Database Management Systems (DBMS): Skilled in designing, implementing, and managing relational databases.
- Graph Representation Learning : Recommender Systems Using GRL, Sequential Recommender Systems
- Programming: Python, PyTorch, TensorFlow, Pandas, Numpy, Matplotlib
- Data Science: Machine learning, Predictive Modeling, Statistical analysis, data visualization
- Tools: Git, Microsoft Office, Google Cloud, Jupyter Notebooks,

Research and Academic:Experimental design and implementation, Academic writing and publication.

- Cross-cultural collaborative research with HausaNLP group.

Ethical AI and Responsible Development:

- Bias detection and mitigation in Educational Data Mining.

Project Management and Leadership:Team leadership and mentoring.

- Team leadership and mentoring.
- Cross-cultural communication and collaboration.

Supervision

2024-Date	**MICHAEL CHIKA ASAA**, Bias Mitigation in Machine Learning Models For Predicting Academic Performance: Ethical Considerations And Practical Solutions	*MS Computer Science*
2024-Date	**HAUWA SANI BELLO**, FAKE NEWS DETECTION IN AFRICAN LANGUAGES (HAUSA):A COMPARATIVE STUDY OF AFRIBERT, ROBERTA, MBERT, XLM-R, AND TINYBERT USING AN IMPROVED DATASET	*MS Computer Science*
2024-Date	**ASHIRU LEKO**, WatchoutNaija: Towards Accurate Detection of Religion-oriented Hate Speech in Nigeria	*MS Computer Science*

Selected Publications

B.A Muhammad, C. Qi, Z. Wu, and **H. K. Ahmad**. An evolving learning style detection approach for online education using bipartite graph embedding. Applied Soft Computing (2024), vol. 152, no. 111230.

H.K. Ahmad, C. Qi, Z. Wu, and B.A. Muhammad. ABiNE-CRS: course recommender system in online education using attributed bipartite network embedding. Applied Intelligence (2022), vol. 53, pp. 4665–4684.

B.A. Muhammad , C. Qi, Z. Wu, and **H. K. Ahmad.** GRL-LS: A learning style detection in online education using graph representation learning. Experts Systems with Applications (2022), vol. 201, no. 117138.

H. K. Ahmad, B. Liu, B. A. Muhammad, and M. Umar. PreBiGE: Course Recommendation Using Course Prerequisite Relation Embedding and Bipartite Graph Embedding. Journal of Networking and Network Applications (2022), vol. 2, no. 4, pp. 161-171.

B.A Muhammad, Bo Liu, **H. K. Ahmad**, M. Umar, and Kenan Qin. GNN-LS: A Learning Style Prediction in Online Environments using Graph Neural Networks. Journal of Networking and Network Applications (2022), vol. 2, no. 4, pp. 172–182.

I. S. Ahmad, **H. K. Ahmad**, S. M. Aliyu, and A. M. Ahmad, "Mathefunic: A Mobile Based Game for Engaging Children with ADHD," International Journal of Perceptive and Cognitive Computing (2020), vol. 6, no. 2, pp. 60–66.

Presentations

Invited Talks

CITAD 2026. *Invited Seminar Presentation*. "The Concept of Readiness to Deploy AI", CITAD Research Methodology Workshop on AI Readiness for Sub-National Governments, Kano, Nigeria.

Contributed Presentations

H. K. Ahmad, B. S. Galadanci, "Design and implementation of mobile learning application: A case study of department of computer science, BUK," A paper presented at Northwest University, Kano 1st international conference, 2015.

Ahmad, H. K., Wu, Z., Muhammad, B. A. (2020). A conceptual framework for top-N recommendation using attributed bipartite graph embedding. In 2020 International Conference on Networking and Network Applications (NaNA) (pp. 400–404). IEEE. https://doi.org/10.1109/nana51271.2020.00074.

Professional Development

Professional Memberships

1. **Member:** Nigerian Women in Information Technology (NIWIIT) 2025 — Date